

## Задание 18 - Корневая файловая система

Выполнил: Шагалиева Анастасия

- 1) Создаем init.c и компилируем программу с -static

```
GNU nano 6.2      ini
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Hello!\n");
    while(1) {
        sleep(1);
    }
    return 0;
}
```

```
svikari@svikari-VirtualBox:~/labs$ sudo nano init.c
svikari@svikari-VirtualBox:~/labs$ arm-linux-gnueabi-gcc -static -o init init.c
```

- 2) Создаем образ корневой файловой системы и запускаем через QEMU

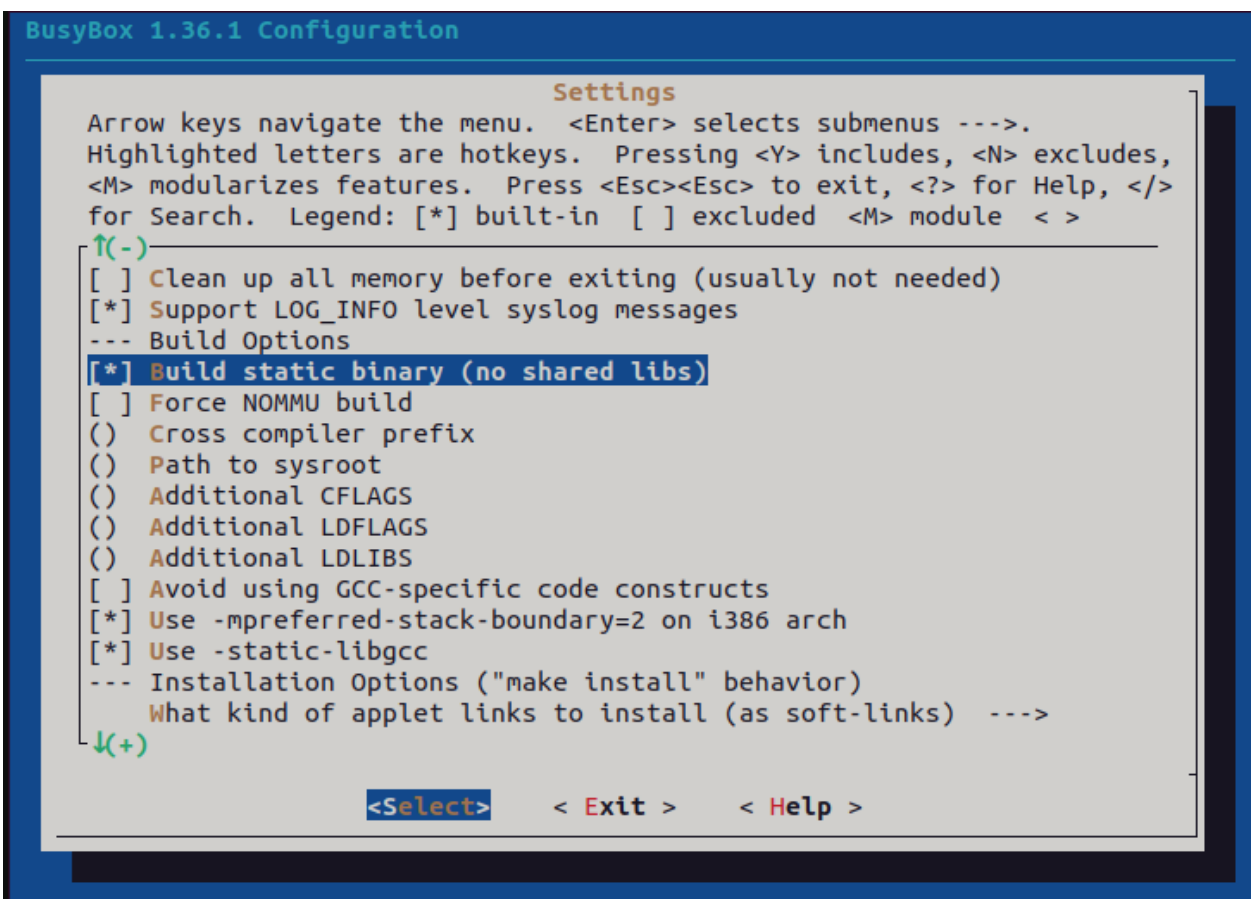
```
svikari@svikari-VirtualBox:~/labs$ echo init | cpio -o -H newc | gzip > initramfs.cpio.gz
1144 blocks
```

```
[ 1.643664] NET: Registered PF_CAN protocol family
[ 1.643704] can: raw protocol
[ 1.643957] can: broadcast manager protocol
[ 1.644053] can: netlink gateway - max_hops=1
[ 1.645315] Key type dns_resolver registered
[ 1.646215] ThumbEE CPU extension supported.
[ 1.646276] Registering SWP/SWPB emulation handler
[ 1.661577] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bu
s@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga
@7,00000000/10006000.kmi/serio0/input/input0
[ 1.685185] Loading compiled-in X.509 certificates
[ 1.701753] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud
= 0) is a PL011 rev1
[ 1.704008] printk: console [ttyAMA0] enabled
[ 1.706399] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud
= 0) is a PL011 rev1
[ 1.707653] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud
= 0) is a PL011 rev1
[ 1.748136] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud
= 0) is a PL011 rev1
[ 1.758133] clk: Disabling unused clocks
[ 1.759810] PM: genpd: Disabling unused power domains
[ 1.789833] Freeing unused kernel image (initmem) memory: 2048K
[ 1.795814] Run /init as init process
Hello!
[ 2.090791] input: ImEXPS/2 Generic Explorer Mouse as /devices/platform/bus@4
0000000/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@40000
000:iofpga@7,00000000/10007000.kmi/serio1/input/input2
```

3) Делаем default config в busybox

```
svikari@svikari-VirtualBox:~/busybox-1.36.1$ ARCH=arm make defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/basic/split-include
HOSTCC scripts/basic/docproc
GEN include/applets.h
GEN include/usage.h
GEN modutils/Kbuild
GEN modutils/Config.in
GEN miscutils/Kbuild
GEN miscutils/Config.in
GEN klibc-utils/Kbuild
GEN klibc-utils/Config.in
GEN console-tools/Kbuild
GEN console-tools/Config.in
GEN applets/Kbuild
GEN findutils/Kbuild
GEN findutils/Config.in
GEN procps/Kbuild
GEN procps/Config.in
```

4) Запускаем ARCH=arm make menuconfig для просмотра меню настроек



5) Собираем корневую файловую систему и запускаем ее

```
svikari@svikari-VirtualBox:~/busybox-1.36.1$ make
CC      applets/applets.o
LD      applets/built-in.o
LD      archival/built-in.o
CC      archival/bbunzip.o
CC      archival/bzip2.o
CC      archival/chksum_and_xwrite_tar_header.o
CC      archival/cpio.o
CC      archival/dpkg.o
CC      archival/dpkg_deb.o
CC      archival/gzip.o
```

```
svikari@svikari-VirtualBox:~/busybox-1.36.1$ make install
./_install//bin/arch -> busybox
./_install//bin/ash -> busybox
./_install//bin/base32 -> busybox
./_install//bin/base64 -> busybox
./_install//bin/cat -> busybox
./_install//bin/chattr -> busybox
./_install//bin/chgrp -> busybox
./_install//bin/chmod -> busybox
./_install//bin/chown -> busybox
./_install//bin/conspy -> busybox
./_install//bin/cp -> busybox
./_install//bin/cpio -> busybox
./_install//bin/cttyhack -> busybox
```

```
svikari@svikari-VirtualBox:~/busybox-1.36.1$ cd _install/
svikari@svikari-VirtualBox:~/busybox-1.36.1/_install$ find . | cpio -o -H newc |
gzip > initramfs.cpio.gz
cpio: File ./initramfs.cpio.gz grew, 1048576 new bytes not copied
7238 blocks
```

```
svikari@svikari-VirtualBox:~/labs$ qemu-system-arm -M vexpress-a9 -kernel zImage
-initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 rd
init=/bin/ash" -nographic
```

```
[ 1.659082] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud
= 0) is a PL011 rev1
[ 1.661042] printk: console [ttyAMA0] enabled
[ 1.663404] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud
= 0) is a PL011 rev1
[ 1.664531] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud
= 0) is a PL011 rev1
[ 1.665508] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud
= 0) is a PL011 rev1
[ 1.695316] clk: Disabling unused clocks
[ 1.695601] PM: genpd: Disabling unused power domains

[ 1.731168] Freeing unused kernel image (initmem) memory: 2048K
[ 1.734291] Run /init as init process
Please press Enter to activate this console. [ 2.091375] input: ImExPS/2 Gene
ric Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bu
s@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/
serio1/input/input2
[ 12.097237] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 12.097305] amba 100e0000.memory-controller: deferred probe pending: (reason
unknown)
[ 12.097315] amba 100e1000.memory-controller: deferred probe pending: (reason
unknown)
[ 12.097321] amba 100e5000.watchdog: deferred probe pending: (reason unknown)

/bin/ash: can't access tty; job control turned off
~ # ls
bin  dev  etc  init  lib  proc  root  sbin  sys  tmp  usr  var
~ #
```